

PHANEENDRA GAVARA

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SUMMARY

M.S. Data Science candidate at ASU (GPA 3.90/4.0) with production experience building LLM pipelines, RAG systems, multi-agent frameworks, cloud-native APIs, and real-time ML applications on **Azure**, **AWS**, and **GCP**. Skilled in Python, PyTorch, OpenAI API, FastAPI, LangChain, and MLOps tooling. Targeting ML Engineer, LLM/GenAI Engineer, AI Product Manager, and Research Engineer roles.

EDUCATION

Arizona State University

M.S. Data Science, Analytics & Engineering **GPA: 3.90 / 4.0**

- Concentration: Computing and Decision Analytics

Tempe, AZ

Aug 2024 – May 2026

IIIT Bhubaneswar

B.Tech in Computer Science Engineering

- Core CS: Algorithms & Data Structures, DBMS, Operating Systems, Computer Networks

Bhubaneswar, India

Dec 2020 – May 2024

TECHNICAL SKILLS

LLM / GenAI: OpenAI API (GPT-4o), Azure OpenAI, LangChain, LangGraph, RAG Pipelines, Prompt Engineering, Fine-Tuning, Autonomous Agents, Semantic Search, Embeddings, SSE Streaming

Vector & Retrieval: Pinecone, Hybrid Search (BM25 + Embeddings), Semantic Re-ranking, Chunking Strategies

ML & Deep Learning: PyTorch, Scikit-learn, HuggingFace Transformers, DistilBERT, CNNs, NLP, Audio ML (Librosa), SARIMAX

MLOps & Deployment: FastAPI, JWT/RBAC, MLflow, Docker, CI/CD, REST APIs, Model Monitoring, Experiment Tracking

Cloud - AWS: S3, EC2, SageMaker, Lambda; **Azure:** SQL Database, App Service, Blob Storage, AI Search, Key Vault, App Insights; **GCP:** Vertex AI, BigQuery

Languages & Stack: Python, TypeScript, JavaScript, SQL, Next.js, React, Bicep IaC

AI PM & Research: System Design, Product Roadmapping, LLM Eval Frameworks, A/B Testing

PROJECTS

AI-Powered Portfolio + Sparky | *Next.js 14, GPT-4o, Pinecone, RAG, TypeScript, Vercel* | [Live](#)

May 2026

- Shipped production RAG pipeline (GPT-4o + Pinecone vector search) powering Sparky - an AI assistant that streams context-aware answers about projects and background via SSE
- Architected full LLM product: embedding pipeline, chunking strategy, retrieval evaluation, and 9-theme UI; open-source — demonstrates end-to-end AI PM + engineer ownership

InsightHub - AI Business Analytics Platform | *Python, FastAPI, Azure OpenAI, Azure SQL, React* | [Live](#)

May 2026

- Hybrid RAG search over 20 business documents (BM25 + ada-002 + Azure AI Search semantic re-ranking); GPT-4o in JSON mode generates structured insights from live SQL metrics across 4 business categories
- FastAPI backend with JWT auth + RBAC (Admin/Analyst/Viewer), 14 endpoints, Application Insights telemetry; Bicep IaC for all 13 Azure resources; React 18 dashboard with 6 pages deployed on Vercel

ASU LLM Evaluation Pipeline | *Python, LangChain, RAGAS, GPT-4o, Pinecone, GitHub Actions, React* | [GitHub](#)

May 2026

- Automated CI/CD pipeline evaluating a university RAG chatbot on every push: GPT-4o-mini judges 100 test questions for faithfulness, relevancy, and hallucination rate — 25× cheaper than GPT-4o at \$0.87/run vs \$21.75
- Enforced 6 automated quality gates (faithfulness ≥ 0.80 , hallucination $\leq 10\%$, latency p95 $\leq 15s$) blocking CI on regression; results auto-committed with [skip ci] and visualised on a live React dashboard (Vercel, 30s auto-refresh)

Republic of Bean - Multi-Agent AI Parliament | *Python, OpenAI GPT-4, LangGraph, Next.js*

Apr 2025

- Multi-agent framework with autonomous GPT-4 agents holding distinct ideological personas that debate policy, vote, and generate structured outcome reports — showcases persona engineering and LangGraph orchestration

Industrial AI Copilot | *Python, AWS Lambda, AWS S3, Next.js, SWR, Vercel*

Mar 2026

- Production real-time anomaly detection across 4 sensor streams (sub-second alerting); automated incident lifecycle: audio alert → work order → technician email; deployed on Vercel with AWS Lambda event triggers and S3 telemetry archival

AI Voice Turing Test | *Python, Librosa, HuggingFace Transformers, NLP* | **3rd Place - ASU Hackathon**

Oct 2024

- MFCCs + spectral feature extraction combined with NLP sentiment analysis to classify human vs. robotic calls; audited demographic bias in model predictions and built interactive real-time classification demo

EXPERIENCE

Teaching Assistant & Grader

Aug 2025 – May 2026

Arizona State University - Fulton Schools of Engineering

Tempe, AZ

- Supported instruction across 4 courses: CSE 540 (Blockchain), CSE 230 (Assembly Language), CSE 180 (Computer Literacy), ASU 101; graded technical coursework and mentored students

Executive Member (Founding Member)

Aug 2021 – Dec 2024

Help N Hope Charitable Society (NGO)

Andhra Pradesh, India

- Co-founded nonprofit focused on accessible education; appointed Executive Member in March 2024